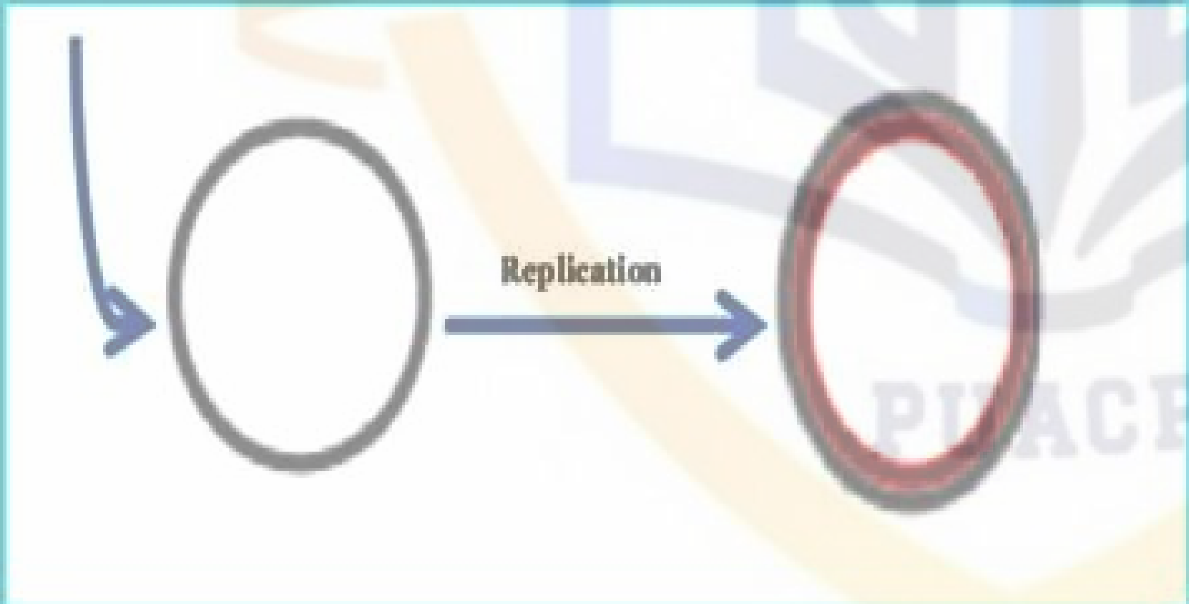
ROLLING CIRCLE OF REPLICATION



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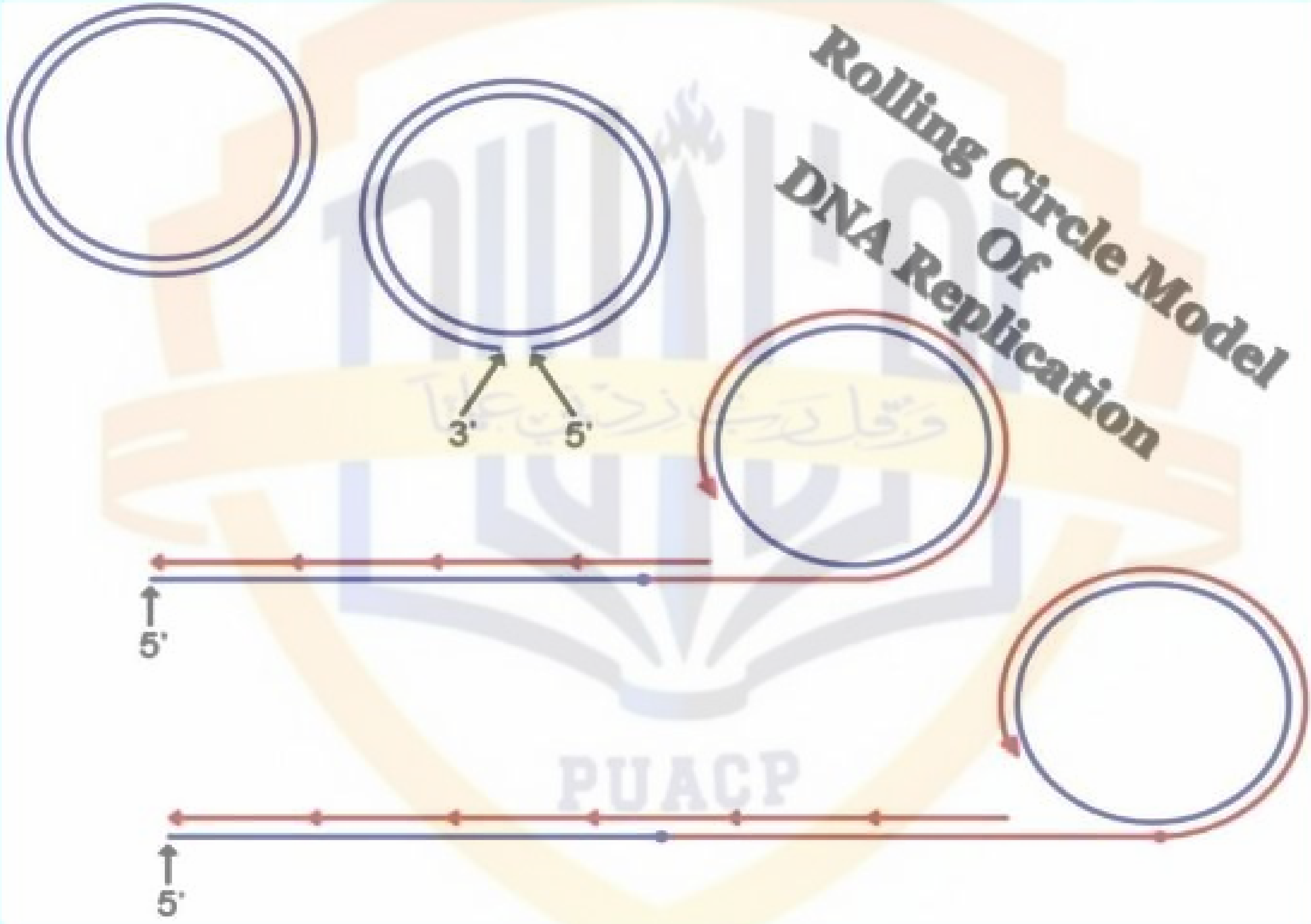
Rolling Circle Model Of DNA Replication



INTRODUCTION:

*Rolling circle replication is a process of unidirectional nucleic acid replication.

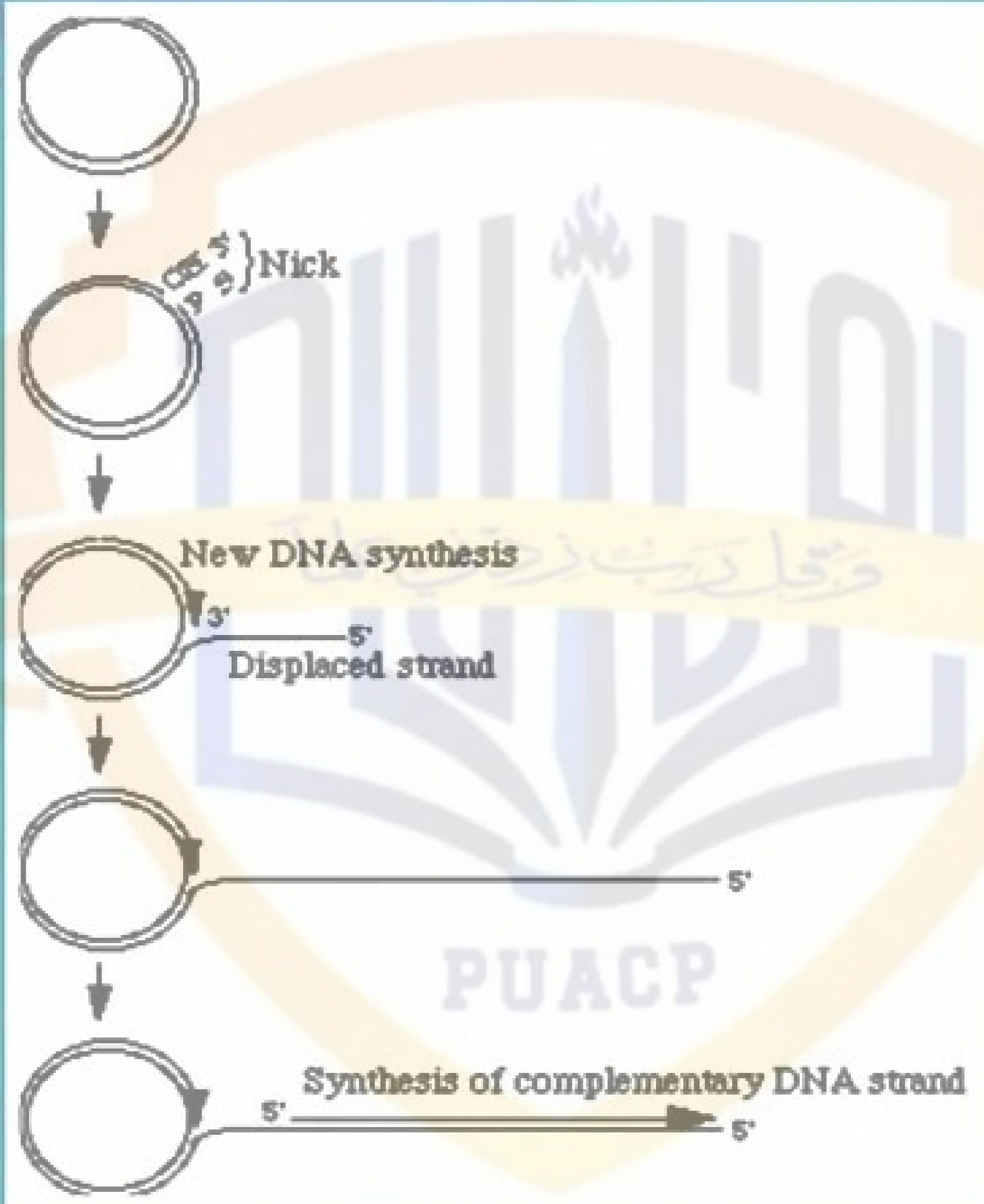
* can rapidly synthesize multiple copies of circular molecules of DNA or RNA, such as plasmids.

* Eucaryotic also replicate.

* widely used in molecular biology & biomedical nanotechnology, especially in the field of biosensing (as a method of signal Amplification).

STEPS:

1) Circular ds DNA will be "nicked"
2) 3` end is elongated →Leading strand
3) 5` end displaced → Lagging strand
   made up of double stranded by OKAZAKI
   fragments.
4) Replication of both " unnicked" and displaced ss
   DNA
5) Displaced DNA circulates and synthesis its own
   complementary strand.



Nick
New DNA synthesis
3'
5'
Displaced strand
5'
Synthesis of complementary DNA strand
5'
5'

→ Replication in eucaryotes is bidirectional, this type is Unidirectional.

→Ideal example of this type is the circular plasmid is bacteria, as it happens only in circular genomes.

Initiation

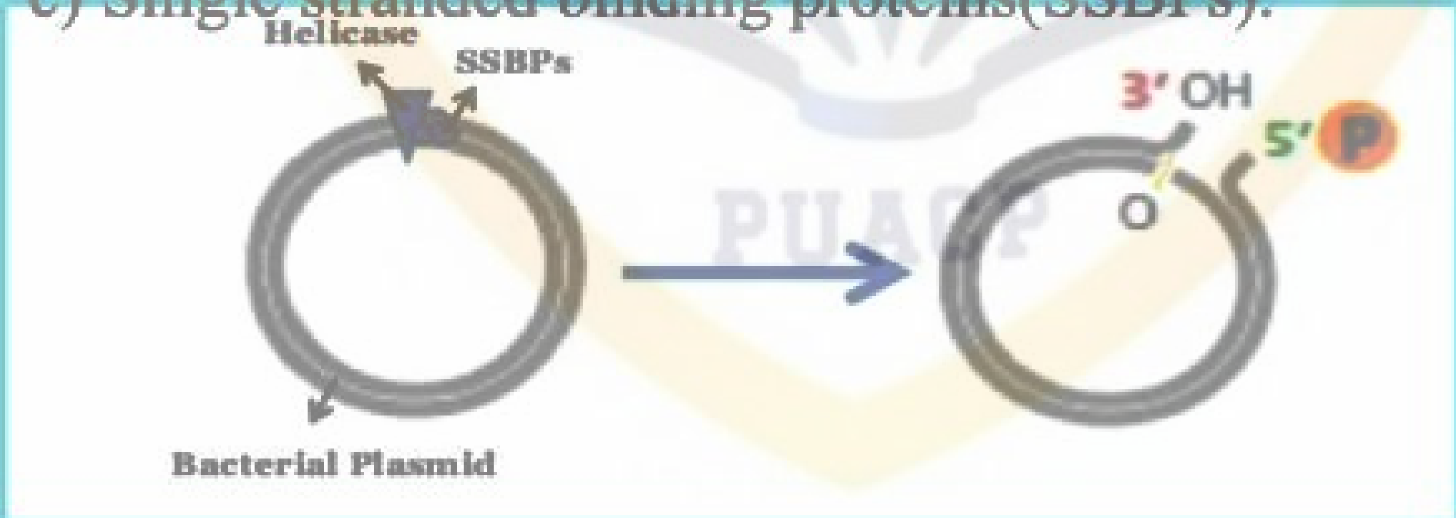
Initiates

- phosphate ends, by the action of:

a) Helicase

b) Topoisomerases

c) Single stranded binding proteins(SSBPs).

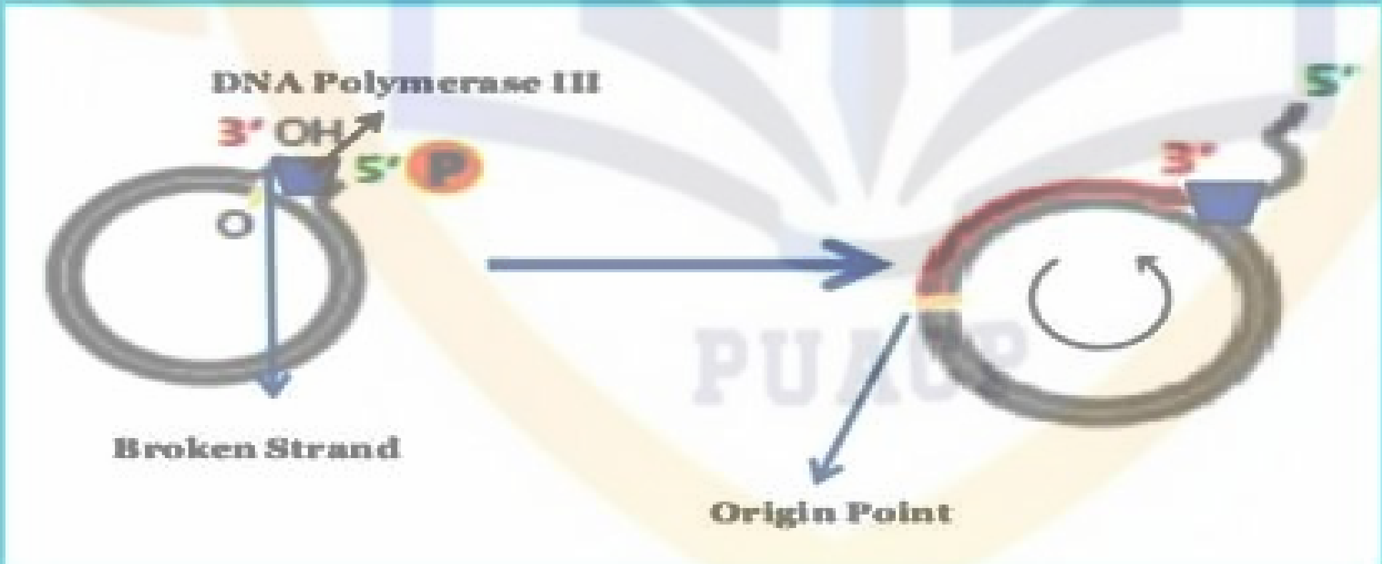


Elongation

→ **For Elongation,**

- OH group of broken strand, using the unbroken strand as a template. The polymerase will start to move in a circle for elongation, due to which it is named as Rolling Circle Model.

end will be displaced and will grow out like a waving thread.

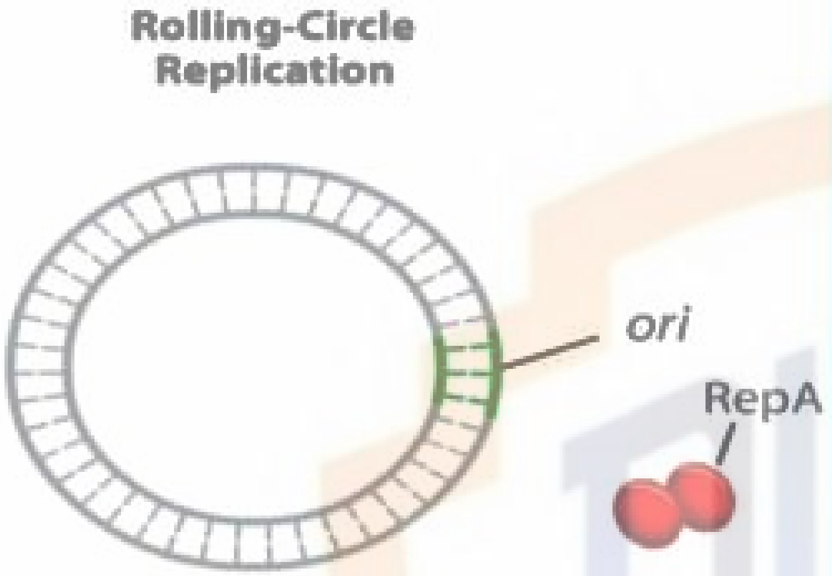


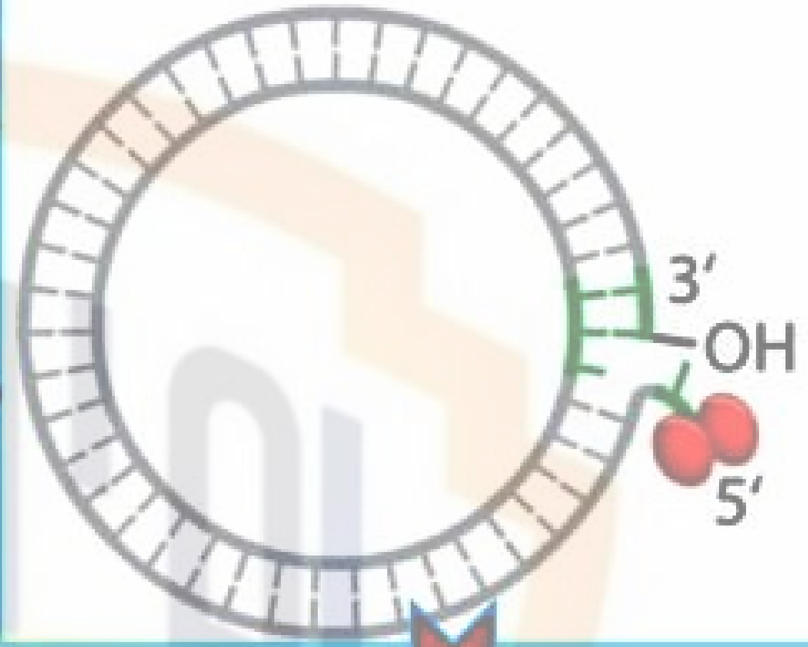
Termination

* At the point of termination, the linear DNA molecule is cleaved from the circle resulting in a double stranded circular DNA molecule and a single- stranded linear DNA molecule.

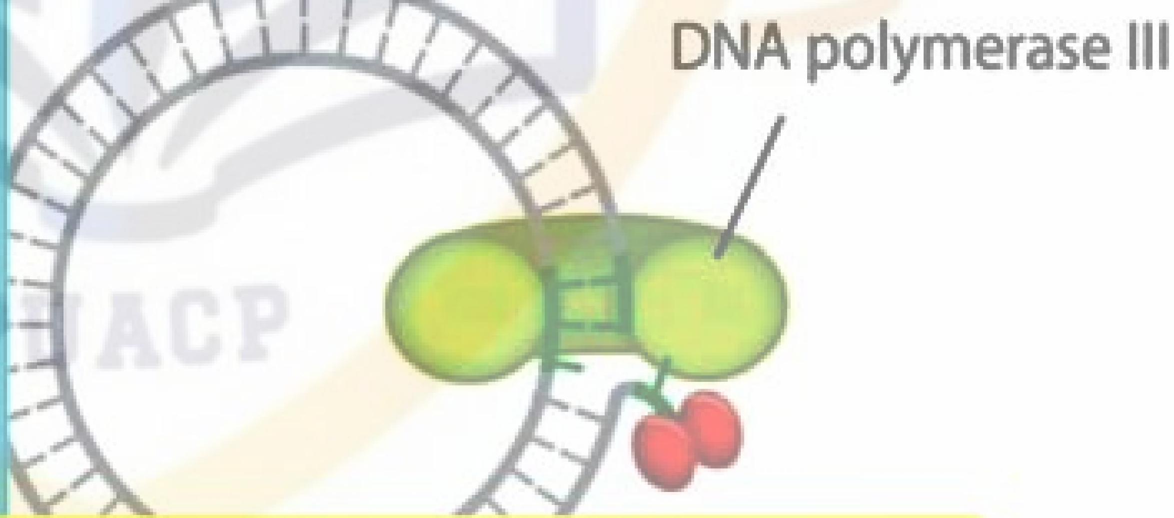
* The linear single stranded molecule is circularized by the action of ligase and then replication to double stranded circular plasmid molecule.

Diagrammatic representation of Rolling circle

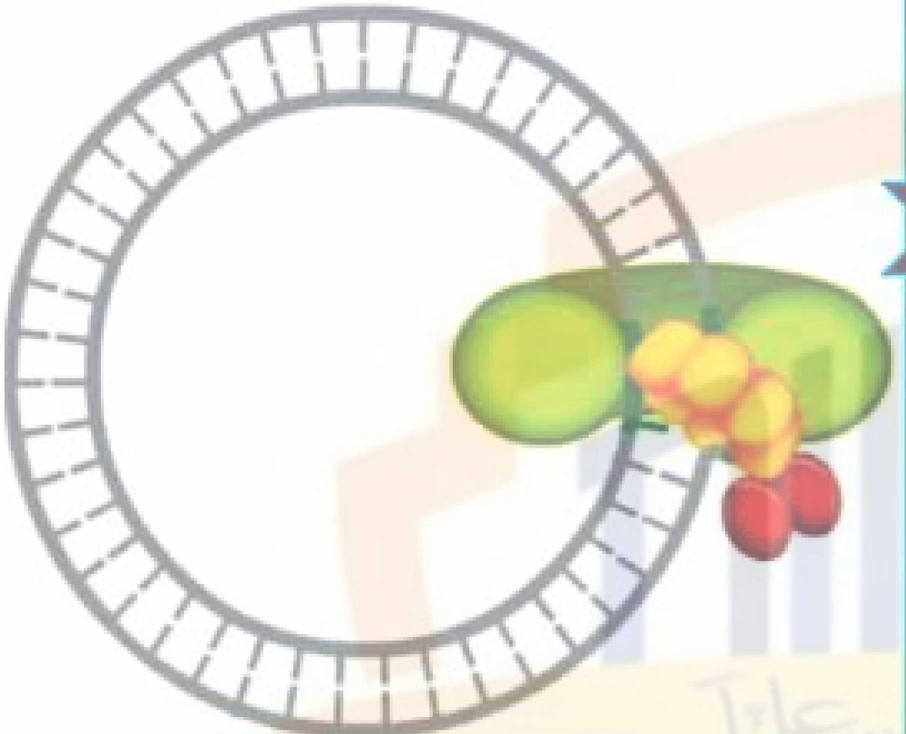




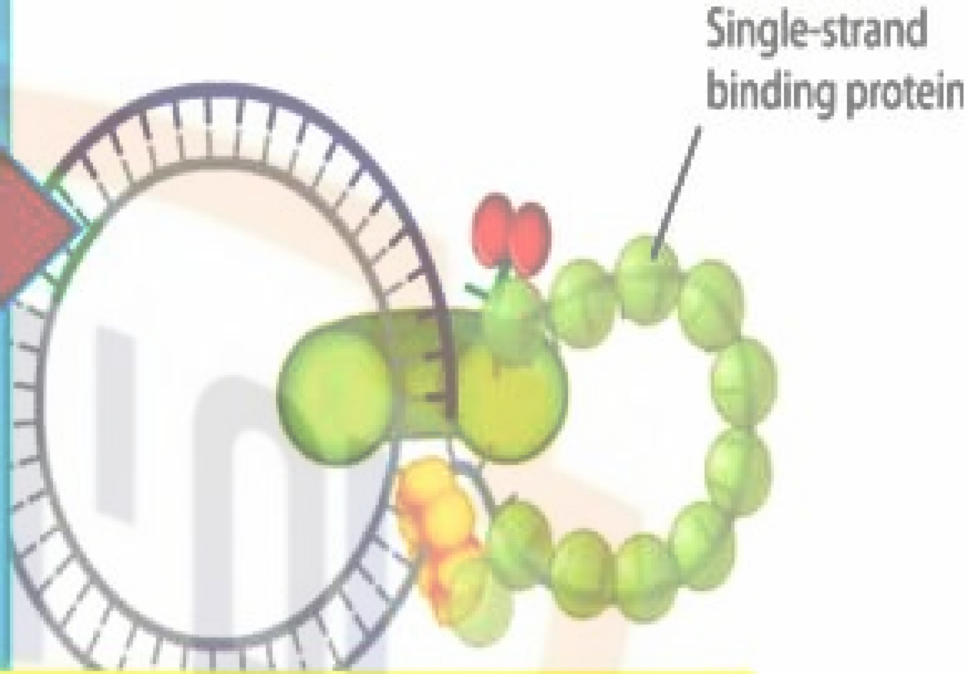
Replication starts at ORI region by a initiator Rep A



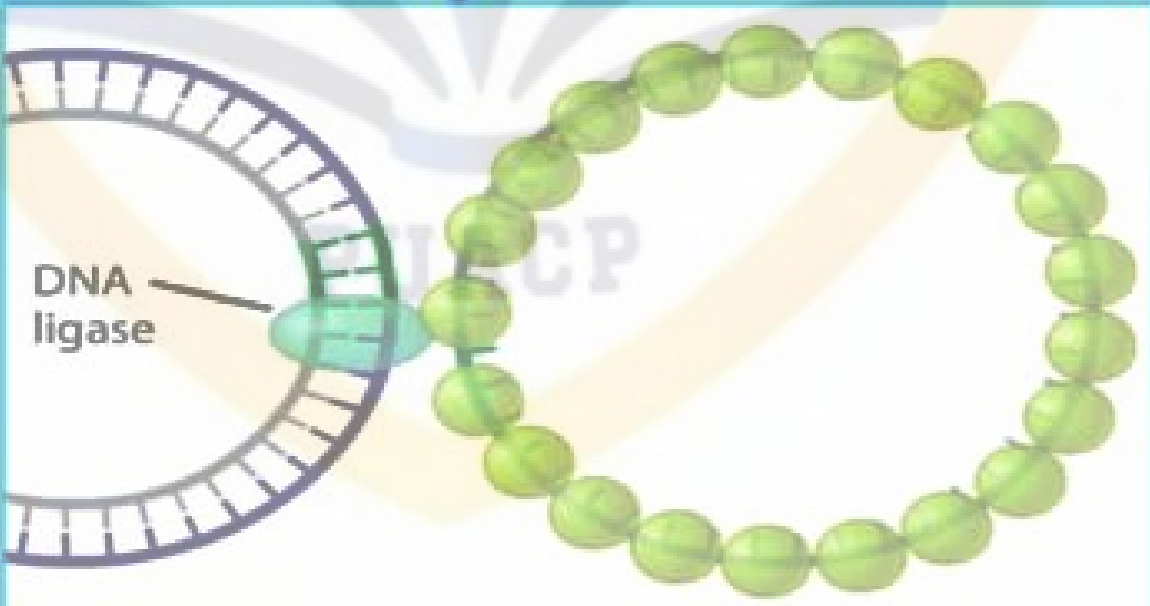
Region binds with DNA Polymerase and replication starts.



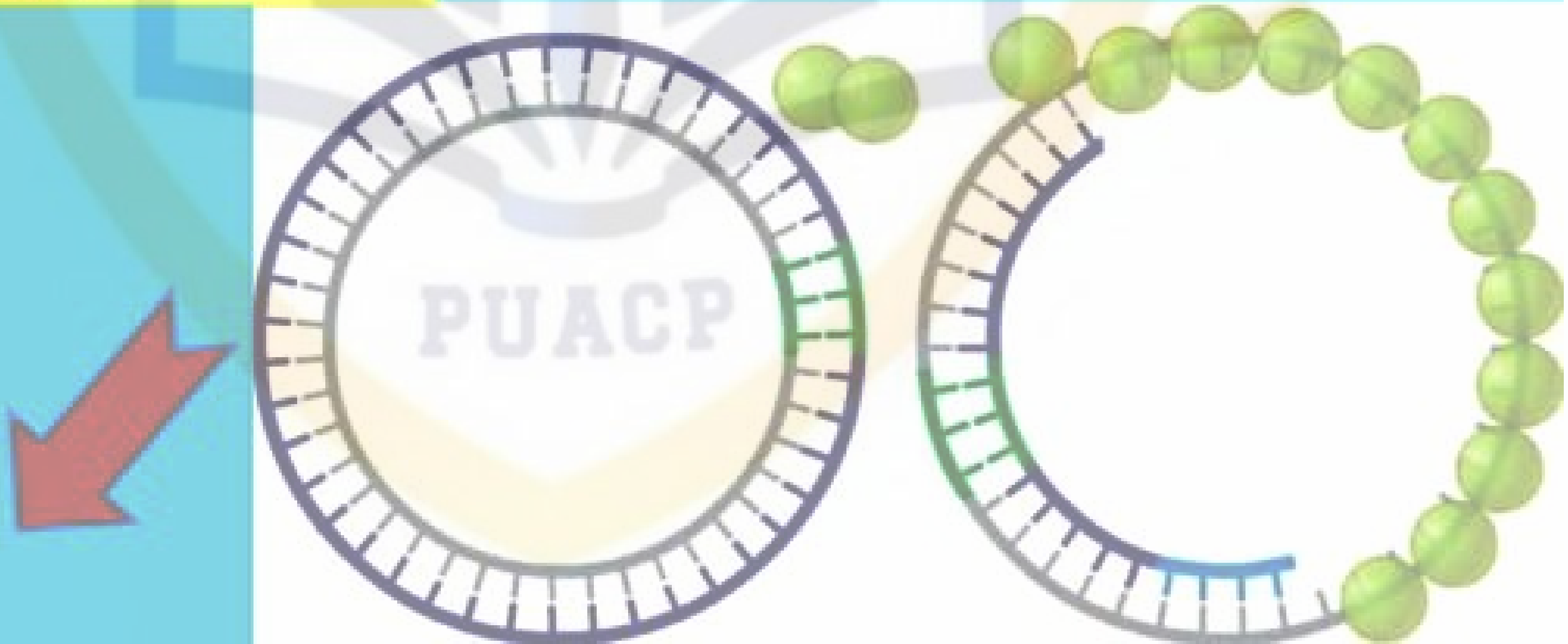
Helicase also attaches to the region

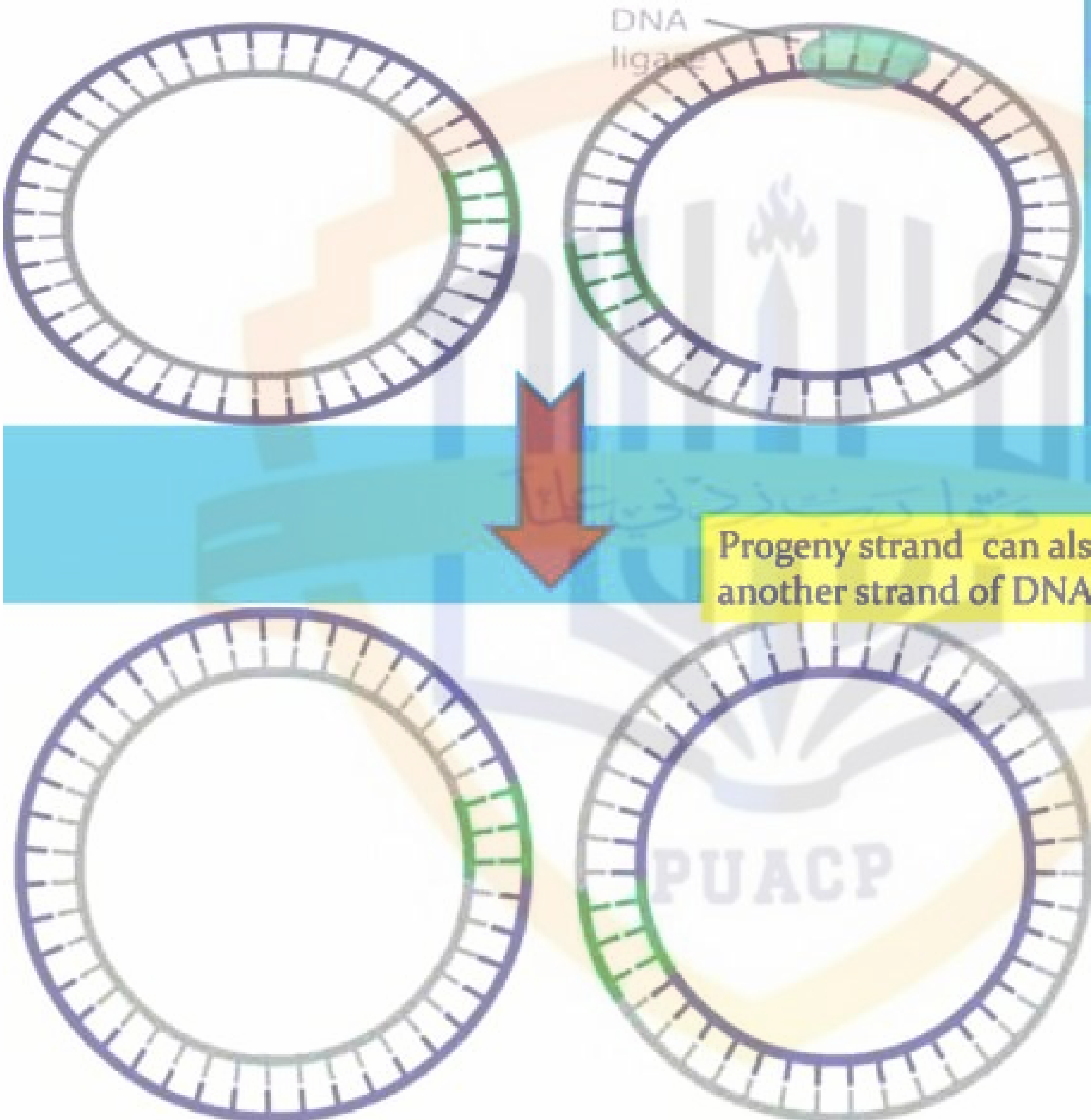


Progeny positive strands synthesis



Positive strands also synthesis its
complementary synthesis

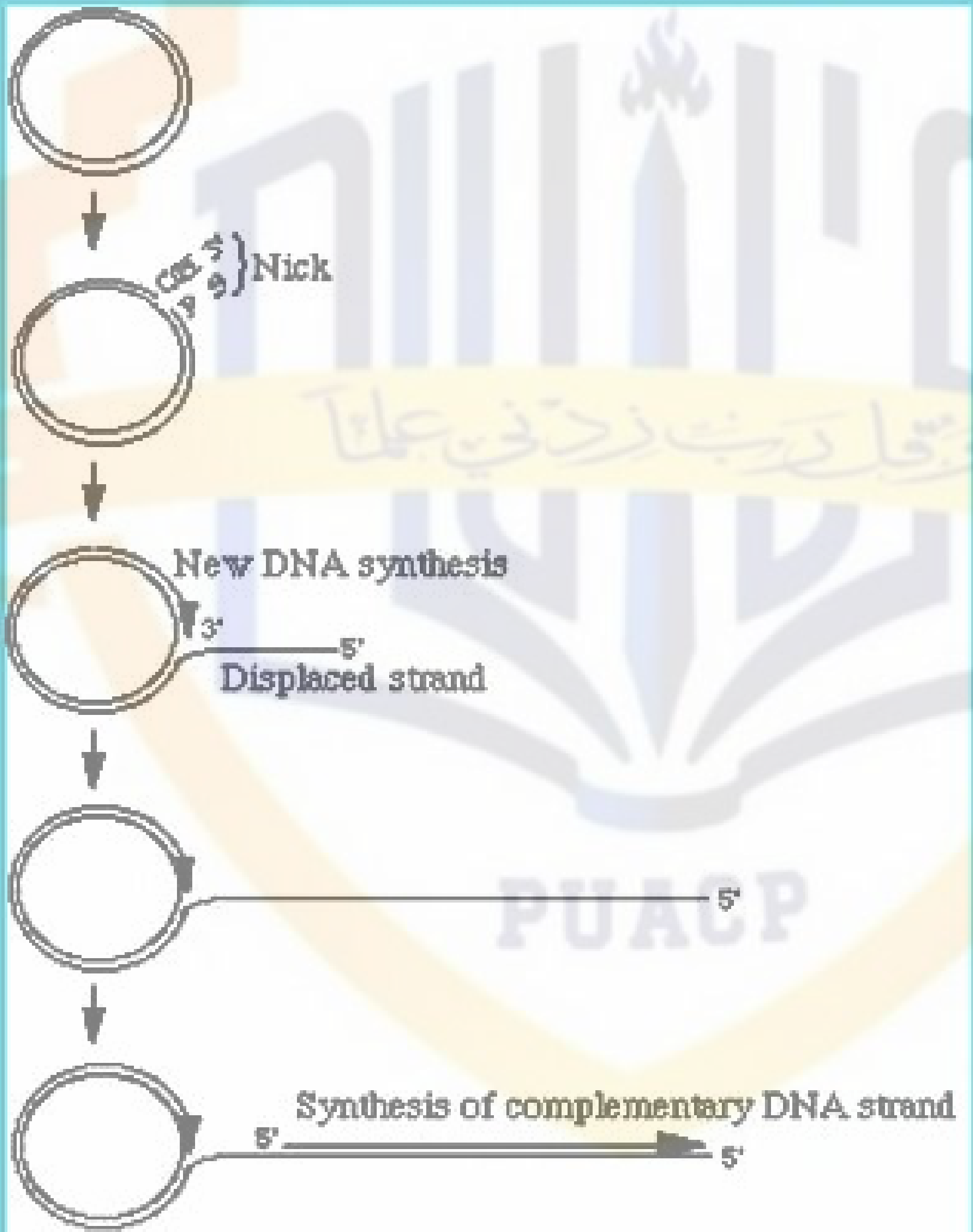




Breakage is joining by Ligase

Progeny strand can also synthesis another strand of DNA

Overall explanation of Rolling Circle:



Example:

Viral DNA

* Human herpes virus
* Human papilloma virus
* Geminivirus

Viral RNA

* pospiviridiae
* Avsunviridiae